

Problem Statement 2

E-Commerce Product Catalog Screen

Background

You are part of a frontend team building an E-Commerce Web Application.

The backend product APIs are still under development, but the UI team needs to complete the Product Catalog screen.

To continue development without waiting for the backend, the team decides to use a fake public REST API from Fake Store API.

Objective

Develop a React application that fetches product data from a fake API and displays it in a Product Listing page.

Functional Requirements

API Details

Method: GET

API URL:

<https://fakestoreapi.com/products>

Fields to Display

Product Image

Product Title

Product Price

Product Category

UI Requirements

Display products in card layout

Show product image, title, price, and category

Show Loading indicator while fetching data

Show Error message if API call fails

Fetch data only once on component load

Technical Constraints

React Functional Components

Axios for API calls

useState and useEffect

Expected Solution (High Level)

Create an API service using Axios

Call the API inside useEffect

Store products in state

Render product cards dynamically

Handle loading and error states

Learning Outcomes

- ✓ API integration in React
- ✓ Handling asynchronous calls
- ✓ Rendering lists dynamically
- ✓ Real-world e-commerce scenario

Problem Statement 3

Task Management Dashboard

Background

You are working on a Task Management System used by internal teams to track daily work.

The backend team is not ready with real APIs yet, so the frontend team decides to mock task data using JSONPlaceholder.

Objective

Build a Task Dashboard that fetches and displays tasks assigned to users.

Functional Requirements

API Details

Method: GET

API URL:

<https://jsonplaceholder.typicode.com/todos>

Fields to Display

Task Title

Task Status (Completed / Pending)

User ID

UI Requirements

Display tasks in a list or table

Show task status clearly (Completed or Pending)

Display Loading message during API call

Display Error message on failure

Fetch tasks only once when component loads

Technical Constraints

React Functional Components

Axios for API calls

useEffect & useState

Expected Solution (High Level)

Create Axios service for tasks

Fetch tasks using useEffect

Map API response to UI

Use conditional rendering for task status

Learning Outcomes

- ✓ Consuming REST APIs
- ✓ Conditional rendering in React
- ✓ State management basics
- ✓ Enterprise-style dashboard scenario